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1.Requirement:

Design, supply, installation and commissioning of a fully automatic Bottle Syrup filling line which is capable of 15,000 liters / day or 25,000 bottles / day @ 600ml / bottle with all accessories. Output speed is 60 Bottle/Min.

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2. Technical Inputs:

As per Attached Annexure II. and Make list.

3. Scope of Work

Scope of work shall be the complete Detailed Engineering, Procurement of Materials, Manufacturing, testing & supply of fully automatic bottle syrup filling line with required accessories.

The scope of supply includes but not limited to the followings-

1.1 Design of fully automatic bottle syrup filling line as per NIHVA specification sheet (Annexure II) & submit design & GA drawing for approval.
1.2 Procurement of all material required for fully automatic bottle syrup filling line.
1.3 Establishment of Quality Assurance Plan & submit for NIHVA approval.
1.4 Procurement & Fabrication as per finalized Quality Assurance Plan.
1.5 Trial of complete fully automatic bottle syrup filling line.
1.6 Supply of complete fully automatic bottle syrup filling line at site.
1.7 Submission of all the As-built drawings, Data sheets & QAP adherence with necessary documents, manual of system for NIHVA final records.
1.8 Submission of Project Schedule, Weekly Progress Report.

Note: Maximum utilization of modern technologies after NIHVA approval.

2. Special requirements:

Bearing:

2.1.1 All bearings and mounting should be SKF make only.
2.1.2 A common greasing position for the all bearings to be provided.

Paint

2.1.3 The paint specification shall be compatible with the external environment consisting of fine urea dust particles which is very corrosive and hygroscopic in nature.
2.1.4 Paint Ral Code require NIHVA approval.

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Design Data:

1.1.1 2D and 3D Design soft data should be share to NIHVA.
1.1.2 Detail BOM with source should be share to NIHVA.

Ducting:

1.1.3 Ducting of Fully automatic bottle syrup filling line is Hinge Type.
1.1.4 There is separate duct for electric cable and pneumatic piping
1.1.5 All electrical cable and pneumatic pipe should be proper routing with good aesthetic.

Manual:

1.1.6 Manual of fully automatic bottle filling line is submitted to NIHVA before dispatch.
1.1.7 PLC program should be submitted to NIHVA before dispatch.
1.1.8 All Spare required to fully automatic bottle syrup filling line in future shall be mention in Manual.
1.1.9 Critical Spare part List should provide separately.
1.1.10 OEM certificate should submit to NIHVA.

Safety:

1.1.1 Maximum safety Equipment consider in Line.
1.1.2 All Motor and Gear box is proper covering.
1.1.3 Pinch point in conveyor and line should be cover by metallic cover.
1.1.4 Direction, flow indication is mention in Line.
1.1.5 The vibration limit shall be as per zones A and B of ISO 10816-1
1.1.6 Noise levels produced by any rotating equipment individually and collectively shall not exceed 85 dB (A) measured at distance of 1.5 MT from the source in any direction under any load condition.
1.1.7 All the Fasteners shall be as per ASTM A 193 B7/ASTM A 194B8 with two washers.
1.1.8 The vibration limit shall be as per zones A and B of ISO 10816-1.
1.1.9 Poka-yoke should provide for Automatic bottle syrup filling line.

Motor and Gear Box:

1.1.1 Motor Insulation must be at least F class
1.1.2 Efficiency Class: IE2/IE3.
1.1.3 All Motor and Gear box is proper covering.

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PLC and HMI:

1.1.1 PLC program should be submitted to NIHVA before dispatch.
1.1.2 HMI program should design in such a way that they show indivisible station performance.
1.1.3 HMI program should design in such a way fault finding of all station is easy.
1.1.4 Integration of all station with help of PLC and HMI is at BN enterprises scope.
1.1.5 HMI screen size is 14inch

Air Unit:

1.1.1 Air preparation Unit (APU) is compulsory for conveyor line.
1.1.2 FRL (Filter Regulator and Lubricant) should provide with proper identification color mark.

Spare Part:

1.1.1 All Spare required to fully automatic bottle syrup filling line in future shall be mention in Manual.
1.1.2 Critical Spare part List should provide separately with price.

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2. Minimum Inspection and Test Requirements:

List of Test Report		
Sr No	**Document**	**Internal/External**
1	Surface roughness Test	External
2	Welding Test	External
3	Tank leakage Test	External
4	Raw material Test Certificate	External
5	Final inspection Assembly- Visual, dimensional	Internal
6	Noise Report (less than 85db)	Internal
7	Motor warranty card	External
8	Performance Repeatability Report for each station	Internal
9	Output min 60 bottle/Min	Internal

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3. **Expected Completion Time**

The delivery of above assignment shall be in a 6 weeks' time frame from the date of award of contract. Delayed in dispatch cause to penalty charges of 0.05% for a week or 5% of overall Po valve whichever is low.

4. **Applicable Engineering Codes & Standards:**

It is the primary responsibility of vendor to design & construct all items related to this contract in accordance with the latest industry standard & codes, including all supplements and addendums issued at time of the order.

5. **Guarantee/Warrantee:**

5.1 BN Enterprises shall guarantee all components related to this contract including spare materials against faulty design, improper material of construction and poor workmanship in addition to performance guarantee. Approval or reviewing by NIHVA for design calculation and detailed shop drawing, will not in any way absolve the BN Enterprises from his responsibility. Should any repair or replacement be necessary owing to any type of failure on account of design, material and workmanship of the item, the BN Enterprises shall in view of this guarantee be bound to replace the same either in part or whole without any time & cost implication to the purchaser. Repaired or replaced part shall also be covered by same guarantee as in case of main supply.

5.2 Guarantee/warrantee shall be valid for 24 months from date of dispatch or 18 months from date of commissioning whichever is earlier.

6. **Quality Assurance & Quality Control:**

6.1 Quality Assurance (QA) shall mean the organization set up, procedures as well as test methods and facilities developed by the vendor/supplier in order to assure that all equipment / items leaving vendor's shop are of the highest possible quality that is either equal to or better than the requirements specified.

6.2 Quality Control (QC), shall mean all the tests, measurements, checks and calibration which are to be carried out at shop in order to compare the actual characteristics of the equipment/unit/system with the specified ones, along with furnishing of the relevant documentation (certificates/ records) containing the data or result of these activities.

6.3 BN Enterprises shall submit a comprehensive description (manual) of QA/QC measures contemplated by them for implementation regarding this specification. It is contractual obligation of the vendor to develop and implement adequate QA/QC systems.

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6.4 QA/QC system shall cover all products and services of the contract i.e. documentation material, shop and site fabrication, transportation and site works, including job subcontracted by the vendor.

6.5 BN enterprise shall permit the NIHVA appointed Quality personal at their premises for inspection. NIHVA shall also communicate the visit plan of their Nominated Quality cum expediter person.

7. Enclosures

7.1 Technical Spec format Annexure II

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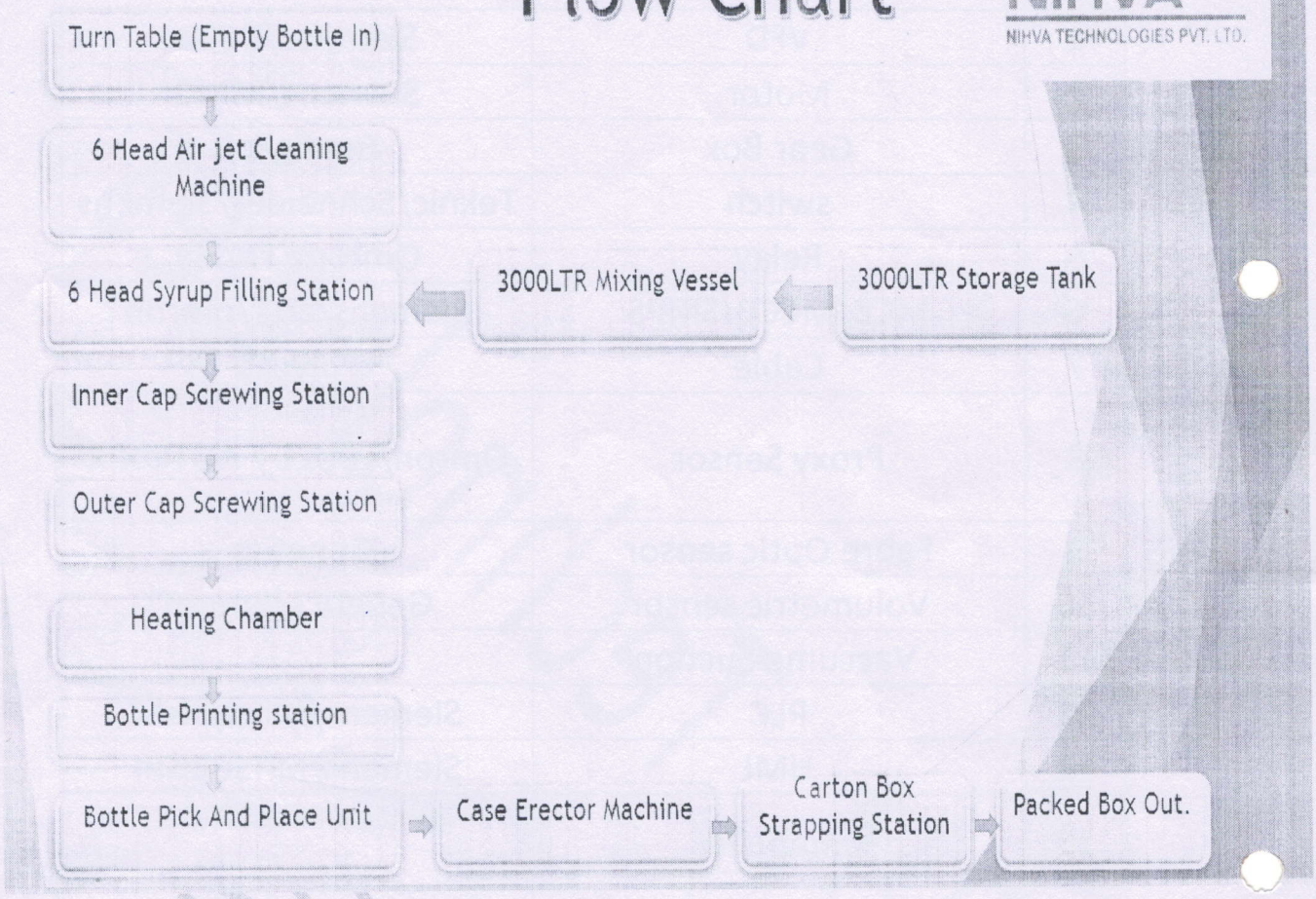
Mandatory Make list		
Sr No	**Details**	**Make**
1	VFD	Siemens/Delta
2	Motor	Siemens/MMC
3	Gear Box	Bonfiglioli
4	switch	Teknic/Schneider/Siemens
5	Relay	Omron /Trinity
6	MCB/MCCB/SMPS	Siemens/L&T/omron
7	Cable	Polycab
8	Proxy Sensor	Omron/LEUZE / AUTONIX
9	Fabre Optic sensor	Genetics
10	Volumetric sensor	Genetics/omron
11	Vaccume Suction	
11	PLC	Siemens/Mitsubishi
12	HMI	Siemens/MITSUBISHI
13	Immpeller	Omron/SELECTRON
14	Hot blower	AC Motor ISO 9000 (MMC MAKE)
15	Pneumatic Cylinder	SMC/ Genetics
16	Pneumatic Fittings	SMC/Genetics
17	Pneumatic Coil	SMC/ Genetics
18	Printer	M6 Handjet Expiry Date Inkjet Printer, TIJ 2.5 Thermal foaming nozzle
19	Water Pipe Line Accessories	Finolex

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Flow Chart

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